

KAJAANI, Finland

WMO: 028970

Lat: 64.283N

Lon: 27.673E

Elev: 440

StdP: 14.46

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-22.5	-16.6	-28.5	1.1	-22.6	-21.8	1.7	-16.4	22.2	28.5	19.5	26.7	0.7	80	0.658

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.7	78.3	64.0	74.8	62.4	71.5	60.7	66.5	74.3	64.2	71.3	62.3	69.1	7.2	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
63.4	88.7	70.1	61.2	82.2	67.8	59.2	76.3	65.8	31.3	74.7	29.6	70.9	28.2	69.1	76.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.4	16.1	14.1	-27.2	82.1	6.5	4.4	-31.8	85.3	-35.6	87.8	-39.2	90.3	-43.9	93.5		
(5)			-27.4	69.0	6.4	3.2	-32.0	71.3	-35.8	73.2	-39.4	75.0	-44.0	77.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	36.1	13.4	13.4	21.8	33.9	45.2	54.9	60.7	56.7	48.0	36.9	27.4	19.6		
	DBStd	18.97	14.45	14.43	10.51	7.94	8.16	6.64	5.93	6.03	6.33	7.38	9.34	13.38		
	HDD50	5893	1134	1026	875	485	195	25	2	12	112	406	679	942		
	HDD65	10588	1599	1446	1340	934	616	310	160	265	512	870	1129	1407		
	CDD50	825	0	0	0	1	45	173	334	220	51	1	0	0		
	CDD65	43	0	0	0	0	1	8	26	8	0	0	0	0		
	CDH74	361	0	0	0	0	23	74	206	58	0	0	0	0		
	CDH80	44	0	0	0	0	1	8	30	5	0	0	0	0		
	WSAvg	6.3	5.4	5.9	6.4	6.3	6.9	6.7	5.8	5.7	6.4	6.8	6.7	6.1		
	PrecAvg	22.0	1.3	1.1	1.1	1.1	1.7	2.2	2.7	3.2	2.3	1.9	1.7	1.4		
(15) Precipitation	PrecMax	28.5	2.4	2.4	2.6	2.6	4.4	5.3	6.7	6.0	4.3	3.9	3.4	2.8		
	PrecMin	16.5	0.6	0.3	0.2	0.4	0.0	0.5	0.7	0.6	0.4	0.0	0.4	0.4		
	PrecStd	3.0	0.5	0.5	0.5	0.6	1.0	1.0	1.4	1.3	1.0	0.9	0.7	0.6		
	PrecStd	3.0	0.5	0.5	0.5	0.6	1.0	1.0	1.4	1.3	1.0	0.9	0.7	0.6		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	37.7	39.4	45.2	62.0	77.4	81.0	83.5	79.9	68.6	54.6	47.5	41.0		
		MCWB	36.0	36.4	38.3	50.0	57.7	63.0	66.9	65.9	59.6	51.3	46.1	40.6		
	2%	DB	34.7	36.3	40.6	54.9	71.3	75.5	79.2	75.0	64.0	51.7	44.0	37.8		
		MCWB	33.8	34.6	35.5	45.0	55.9	61.3	65.5	63.9	57.2	49.4	42.7	36.8		
	5%	DB	33.2	33.8	37.4	50.2	65.1	71.4	75.6	71.3	60.6	49.5	40.8	35.7		
		MCWB	32.6	32.7	33.8	41.8	52.4	59.2	63.7	61.8	55.1	47.6	39.9	34.7		
	10%	DB	31.6	31.9	35.1	46.1	60.2	67.9	72.5	68.0	57.8	47.1	38.3	33.8		
		MCWB	31.0	31.0	32.5	39.8	49.8	57.2	61.8	60.2	53.8	45.5	37.4	32.9		
	0.4%	WB	36.1	36.9	39.2	51.0	60.2	66.6	70.2	68.4	60.7	52.3	46.1	40.2		
		MCDB	37.1	38.6	43.5	58.7	71.3	74.5	78.8	76.8	66.3	53.9	47.4	40.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	33.9	34.6	36.3	46.1	57.1	63.7	67.5	64.7	58.1	49.9	42.7	37.0		
		MCDB	34.4	35.9	39.4	52.9	68.2	71.7	75.8	71.7	62.8	51.4	43.7	37.5		
	5%	WB	32.5	32.8	34.5	42.6	53.9	60.9	65.5	62.9	56.0	47.8	39.8	34.7		
		MCDB	33.0	33.7	36.7	48.6	63.8	69.1	73.0	69.4	59.7	49.1	40.6	35.3		
	10%	WB	30.8	30.9	32.8	40.1	50.8	58.4	63.4	61.0	54.3	45.6	37.5	33.0		
		MCDB	31.4	31.8	34.7	45.3	58.7	66.1	70.2	67.4	57.5	47.0	38.2	33.6		
(35) Mean Daily Temperature Range	5% DB	MDBR	12.2	14.1	16.1	16.1	17.7	16.8	16.7	16.0	13.0	8.5	8.1	10.7		
		MCDBR	8.8	9.9	14.9	22.2	27.5	22.7	21.9	21.1	17.0	10.0	8.0	6.8		
		MCWBR	8.7	9.1	11.6	14.1	14.5	12.0	10.9	12.0	11.3	8.1	7.8	6.6		
		MCDBR	9.0	9.5	12.1	20.4	24.4	19.4	18.6	18.6	15.3	9.2	7.8	6.7		
	5% WB	MCWBR	8.9	8.9	9.8	13.7	14.0	10.9	10.2	11.2	10.7	8.1	7.7	6.6		
		MCWBR	8.9	8.9	9.8	13.7	14.0	10.9	10.2	11.2	10.7	8.1	7.7	6.6		
(40) Clear-Sky Solar Irradiance	taub		0.211	0.276	0.305	0.343	0.343	0.344	0.368	0.348	0.325	0.302	0.359	0.205		
	taud		1.956	2.111	2.167	2.284	2.460	2.479	2.432	2.472	2.524	2.383	2.643	1.967		
	Ebn at Noon		108	196	244	263	276	277	266	261	245	193	101	40		
	Edh at Noon		9	20	29	32	30	30	31	27	21	16	8	3		
(44) All-Sky Solar Radiation	RadAvg		45	209	658	1142	1494	1661	1572	1175	649	239	56	16		
	RadStd		5	43	68	122	161	122	158	116	67	39	11	1		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+1.22	N/A	N/A	N/A	N/A	N/A	N/A	-328	N/A	N/A
(47)	Regional (31 neighbors)	+0.99	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page